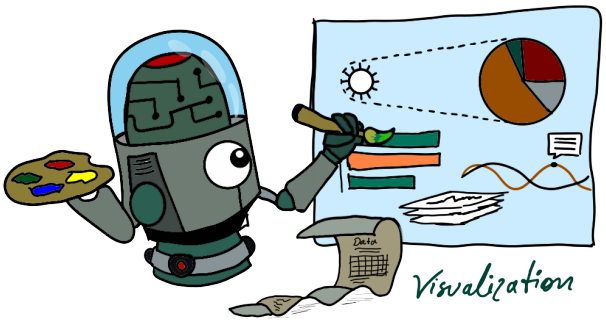
DS5343 – Data Visualization

Universidad de Ingeniería y Tecnología



Prof. Germain Garcia-Zanabria

Data Science Program

Semester 2026–2

Final Project Guidelines

Final submission and exhibition: Wednesday, November 18, 2026

Contents

1 Project Overview **1**
1.1 Project goals . . . 1

2 Topic and Dataset Selection **1**

3 Teams and Project Infrastructure **2**
3.1 Team formation . . . 2
3.2 Repository and collaboration . . . 2
3.3 Required technology . . . 2

4 Project Milestones **3**
4.1 Week 4: Dataset Selection and Submission — September 2, 2026 . . . 3
4.2 Week 5: Team and Project Proposal — September 9, 2026 . . . 4
4.3 Week 6: Data Analysis and Sketches/inspiration — September 16, 2026 . . . 4
4.4 Week 7: Delivery 1 — September 23, 2026 . . . 5
4.5 Week 10: Functional Prototype and Design Review — October 14, 2026 . . . 6
4.6 Week 12: Refined Draft and Peer Critique — October 28, 2026 . . . 7
4.7 Week 15: Final Submission and Exhibition — November 18, 2026 . . . 7

5 Final Deliverables **8**
5.1 Published D3.js application . . . 8
5.2 Conference-style paper . . . 8
5.3 Video teaser . . . 9
5.4 Presentation video . . . 9
5.5 Poster . . . 9
5.6 Project page . . . 9
5.7 Repository, README, and code . . . 9

6 Evaluation Criteria **10**

7 General Quality and Submission Requirements **10**

8 Social Activities Bonus **10**

1 Project Overview

The final project provides a semester-long practical experience in analytical visualization, interaction design, and visual communication using **complex, real-world data**. Each team will select a dataset of its choice, formulate a focused visualization problem, and design, implement, publish, and communicate an interactive visualization solution.

The project is not a machine-learning competition and is not evaluated by the size of the codebase. It is an opportunity to demonstrate mastery in Data Visualization: asking meaningful questions, transforming data appropriately, selecting expressive and effective visual encodings, designing useful interactions, implementing the solution in **D3.js**, deriving defensible insights, and communicating the results as a coherent narrative.

Strong execution is more important than excessive ambition. A focused project with a clear audience, well-justified design choices, reliable implementation, and meaningful findings is preferable to a broad project that attempts to show everything in a dataset.

1.1 Project goals

By the end of the project, each team should be able to:

- define a concrete and well-motivated visualization problem;
- identify the intended users or audience and the decisions or questions the visualization should support;
- assess, clean, transform, and document a real-world dataset;
- formulate a coherent sequence of domain questions that can be answered visually;
- justify visual encodings, interaction techniques, layout, color, and annotation decisions;
- implement an original interactive visualization using D3.js;
- evaluate and refine the design through critique and iteration;
- communicate the project through a published application, paper, videos, poster, project page, and oral presentation.

2 Topic and Dataset Selection

Teams may choose any dataset and topic that they find meaningful. Possible domains include urban data, transportation, public safety, health, sports, entertainment, finance, climate, education, science, social behavior, or another area **approved by the instructor**.

The selected data should be:

- accessible to the entire team and legally usable for the project;
- sufficiently rich to support multiple related analytical questions (multiples tables);
- documented well enough for the team to explain its source, collection process, attributes, and limitations;

- manageable within the semester and compatible with a web-based D3.js implementation;
- preferably temporal, spatial, multivariate, hierarchical, relational, or otherwise structurally interesting.

Before proposing a topic, teams must inspect the actual data. A link to a dataset is not evidence that the data contains the attributes needed to answer the proposed questions. Projects based on private or sensitive data must follow applicable privacy, anonymization, ethical, and access requirements.

3 Teams and Project Infrastructure

3.1 Team formation

Projects will normally be completed in **teams of three students**. Exceptions require explicit authorization from the instructor. All members are expected to contribute throughout the complete process, including problem formulation, data analysis, design, implementation, writing, and presentation.

Each team should agree early on communication practices, meeting frequency, responsibilities, internal deadlines, and procedures for handling delays. Individual contributions must be documented in the repository. The instructor may adjust individual grades when the available evidence indicates substantially unequal participation.

3.2 Repository and collaboration

Each team must maintain a shared GitHub or GitLab repository from the beginning of the project. The repository must contain the source code, data-processing scripts when applicable, documentation, report source, poster, and final submission materials that can be shared legally.

The team is expected to use version control throughout the project. A repository created or populated only near the final deadline will not provide adequate evidence of collaboration or iterative development.

3.3 Required technology

The final interactive visualization must be implemented using **D3.js**. Supporting technologies such as HTML, CSS, JavaScript modules, preprocessing scripts, build tools, or lightweight web frameworks may be used when appropriate, but they must not replace D3.js as the principal visualization implementation technology.

Tools such as Tableau, Power BI, Looker, or Excel may be used privately for early exploration, but they are not acceptable as the final visualization solution. External libraries, code examples, AI-assisted code, datasets, icons, and visual inspirations must be acknowledged. The submitted project must include substantial original work by the team, and every member must be able to explain the implementation and design decisions.

4 Project Milestones

Milestones scaffold the project and provide opportunities for feedback before the final submission. Unless otherwise announced, submissions are due on the specified Wednesday through the course platform. The instructor will communicate the exact submission time and presentation order.

Week	Date	Milestone	Main evidence due
4	Sep. 2	Dataset Selection and Submission	Dataset files, source, license, data dictionary, and access instructions
5	Sep. 9	Team and Project Proposal	Problem, audience, dataset, questions, feasibility, and repository
6	Sep. 16	Data Analysis and Sketches	Processed data samples, transformations, findings, and visualization sketches
7	Sep. 23	Delivery 1	Consolidated first phase, revised scope, design plan, and work plan
10	Oct. 14	Functional Prototype and Design Review	Working D3 prototype, principal interactions, and review presentation
12	Oct. 28	Refined Draft and Peer Critique	Integrated narrative, improved prototype, and documented critique response
15	Nov. 18	Final Submission, Oral Presentation, and Poster Exhibition	Complete final package and public presentation

Week 8 has no project deliverable. This week must remain free of project submissions due the Mid-tem Exam. Teams should nevertheless use the interval to consolidate feedback from Delivery 1 and plan the transition toward implementation.

4.1 Week 4: Dataset Selection and Submission — September 2, 2026

This first milestone is intentionally limited to the data. Teams are not expected to submit a complete project proposal this week. Each team must identify a candidate dataset, inspect its actual contents, and submit a usable data package.

Deliverable: A repository folder named `deliveries/week04/` containing:

- `README.md`: dataset title, source URL, organization or collector, collection method when known, geographic and temporal coverage, access date, license or terms of use, and a short explanation of why the dataset is potentially useful for visualization;
- `data/`: the dataset in CSV format whenever possible;
- `data/sample.csv`: a representative CSV sample when the complete dataset is too large, restricted, obtained through an API, or distributed in another format;
- `data_dictionary.csv`: one row per attribute, including the field name, description, data type, unit or format, possible values when relevant, and known missing-value codes;
- `acquisition.md` or an acquisition script when the complete data must be downloaded, queried through an API, or generated from another source.

Do not upload private, confidential, personally identifiable, copyrighted, or otherwise restricted data to a public repository. In such cases, submit a legally shareable sample and document the access procedure and restrictions.

4.2 Week 5: Team and Project Proposal — September 9, 2026

Submit a concise proposal of approximately 2–3 pages containing:

1. **Team and working title:** team members, proposed title, and repository URL.
2. **Problem statement and background:** the concrete problem, why it matters, and the context a reader needs to understand it.
3. **Audience and purpose:** who should use or read the visualization and what it should help them understand, explore, or decide.
4. **Dataset:** source, collector, collection method when known, relevant attributes, data quality issues, access conditions, and expected preprocessing.
5. **Domain questions:** approximately 5–8 focused questions that form a logical progression and can be answered using the available data.
6. **Initial feasibility:** expected technical challenges, division of work, and principal risks.

Avoid vague goals such as "visualize the dataset" or "find interesting patterns." The proposal should establish a focused analytical direction without prematurely fixing every design decision.

Deliverable: A repository folder named `deliveries/week05/` containing:

- `ProjectProposal.pdf`: the 2–3 page proposal;
- the editable proposal source in `.tex`, `.docx`, or an equivalent source format;
- `PresentationWeek05.pptx` or `PresentationWeek05.pdf`: a concise presentation of the proposed problem, audience, data, and domain questions;
- `README.md`: team members, working title, repository structure, responsibilities, and relevant links;
- the revised Week 4 data package if the dataset changed after feedback.

4.3 Week 6: Data Analysis and Sketches/inspiration — September 16, 2026

For each major domain question, the team must demonstrate that it can derive the necessary information from the dataset and has considered an appropriate visual response.

Deliverable: A short document or project-page update containing:

- a representative sample of the processed data for each major analytical view;
- clear documentation of cleaning, filtering, derivation, aggregation, or transformation steps;

- initial exploratory findings, anomalies, data limitations, and revised questions when necessary;
- sketches for the planned visualizations, including titles, labels, legends, visual encodings, layout, and annotations;
- indications of proposed interactions such as filtering, selection, zooming, linking, details on demand, or animation;
- a short explanation of why each design is appropriate for its question and data type.

Sketches may be hand-drawn or created using design or visualization software. They must be the team's work and detailed enough that another person could understand the intended implementation.

Deliverable: A repository folder named `deliveries/week06/` containing:

- `DataAnalysis.md` : a concise report containing the data understanding, transformations, findings, limitations, and design rationale;
- `PresentationWeek06.pptx` or `PresentationWeek06.pdf`;
- `data/processed/`: the processed CSV files used to support the proposed views;
- `code/`: scripts or notebooks used to clean, transform, aggregate, or analyze the data;
- `sketches/`: sketches exported as PDF, PNG, or another widely accessible format;
- Review at least four papers that provide relevant references or inspiration for the proposed data visualizations, visual encodings, interactions, or sketches. Briefly explain how each paper informs the project's design. (Present the pdfs)
- `README.md`: exact instructions for reproducing the processed data and locating each output.

4.4 Week 7: Delivery 1 — September 23, 2026

Delivery 1 consolidates the first phase of the project. It should include:

- the revised problem statement, audience, and dataset description;
- the revised sequence of domain questions;
- the data analysis and final set of candidate sketches;
- the selected visualization direction and rationale;
- an initial information architecture or layout for the application;
- a prioritized implementation plan for the D3.js solution;
- a division of responsibilities and schedule through Week 15;
- a short reflection explaining what changed after examining the data.

This submission is a scope checkpoint. Teams may be asked to narrow, reframe, or revise the project before continuing.

Deliverable: A repository folder named `deliveries/week07/` containing:

- `Delivery1Report.pdf` or `Delivery1Report.md`: the consolidated first-phase report;
- the editable report source;
- `PresentationWeek07.pptx` or `PresentationWeek07.pdf`;
- the selected sketches and application architecture in PDF or PNG;
- the current data files in CSV format, data dictionary, and processing code;
- an updated root `README.md` with responsibilities, project schedule, repository structure, and reproduction instructions.

4.5 Week 10: Functional Prototype and Design Review — October 14, 2026

Publish a functional D3.js prototype in the repository and, whenever possible, as a web application. The prototype does not need final styling, but it must show substantial progress:

- the principal visualizations must render using the real project data;
- the core interactions must be functional;
- the main analytical flow or navigation must be evident;
- the visual encodings, scales, labels, and legends must be interpretable;
- known limitations and incomplete elements must be documented.

Each team will give a design-review presentation of no more than 5 minutes covering: the problem and audience; a prototype demonstration; principal design decisions and alternatives considered; and specific questions on which the team wants feedback.

Peer feedback should use the structure "I like... / I wish... / What if...?" and should address visual encoding, interaction, usability, analytical value, layout, labeling, and communication. Feedback must be constructive, specific, and feasible.

Deliverable: A repository folder named `deliveries/week10/` containing:

- the complete prototype source code, including HTML, CSS, JavaScript, D3.js modules, and required assets;
- the current processed data in CSV or JSON format;
- `PrototypeReport.pdf` or `PrototypeReport.md`: implemented features, design decisions, known limitations, and questions for reviewers;
- `PresentationWeek10.pptx` or `PresentationWeek10.pdf`;
- a deployed prototype URL when publication is technically and legally possible;
- `README.md` with setup, execution, and deployment instructions.

4.6 Week 12: Refined Draft and Peer Critique — October 28, 2026

Transform the prototype into a coherent visual story or exploratory experience. Submit:

- a refined and published D3.js application with all major views and interactions implemented;
- a clear title, introduction, data description, analytical sequence, findings, and preliminary conclusion;
- improved visual design, including typography, color, annotations, accessibility, responsive layout when feasible, and interaction feedback;
- an initial version of the project page and paper structure;
- a summary of feedback received during the design review;
- a response log stating which suggestions were implemented, deferred, or rejected, with justification;
- the remaining tasks and ownership plan for the final three weeks.

At this point, the project should no longer look like a disconnected collection of charts. A reader unfamiliar with the topic should be able to understand the motivation, move through the visual analysis, and identify the main findings.

Deliverable: A repository folder named `deliveries/week12/` containing:

- the refined application source code and updated deployment link;
- the final-candidate processed data in CSV or JSON format and all reproduction scripts;
- `RefinedDraftReport.pdf` or `RefinedDraftReport.md`: narrative, design rationale, preliminary findings, limitations, and remaining work;
- `PresentationWeek12.pptx` or `PresentationWeek12.pdf`;
- `PeerCritiqueResponse.pdf`: feedback received and the team's response to each substantive suggestion;
- the initial paper source and PDF draft;
- the initial project page and updated `README.md`.

4.7 Week 15: Final Submission and Exhibition — November 18, 2026

All final materials must be complete and accessible from the project repository and project page before the assigned presentation or exhibition time. The team will participate in a poster exhibition and give a 10-minute oral presentation. A short question-and-answer period may follow.

The oral presentation should allocate approximately:

- 2 minutes to the problem, audience, motivation, and data;

- 6 minutes to the visualization, interactions, design decisions, and principal findings;
- 2 minutes to conclusions, limitations, and future work.

Deliverable: The complete final repository and project page, including all artifacts listed in Section 5. The team must also include `PresentationFinal.pptx` or `PresentationFinal.pdf`, the final poster in PDF, the final paper and editable source, the published D3.js application, both video links, the final data package, complete source code, and the final README.

5 Final Deliverables

5.1 Published D3.js application

The final application must be publicly accessible whenever the data and project constraints permit. GitHub Pages is recommended for static HTML/CSS/JavaScript projects. If another hosting service is used, the repository must still contain the source code and complete instructions for reproducing or running the project.

The application must work with the submitted data, load reliably, and provide an experience consistent with its intended audience and purpose. Data sources and relevant external inspirations must be acknowledged in the interface or project documentation.

5.2 Conference-style paper

Submit a 2–4 page paper, excluding references, in a two-column conference format such as IEEE VIS or ACM CHI. The final PDF should be named `FinalPaper.pdf` and accompanied by its editable source.

The paper should contain:

- **GRAPHICAL ABSTRACT:** A graphical representation of the methodology and transformations.
- **Introduction:** problem, audience, motivation, and contributions;
- **Related work:** relevant visualization systems, designs, or research;
- **Data and methods:** data source, transformations, analytical process, and implementation approach;
- **Visualization design:** visual encodings, interactions, layout, and design rationale;
- **Results or case study:** findings and examples showing how the visualization addresses the questions;
- **Discussion:** what was learned, strengths, limitations, and informal evaluation or usage observations;
- **Future work and conclusion:** possible extensions and final takeaways.

5.3 Video teaser

Create a 1-minute teaser that acts as a trailer for the project. Rather than attempting to demonstrate every feature, the teaser should communicate the motivation and highlight one or two compelling findings or interactions that encourage viewers to explore the full application.

5.4 Presentation video

Create a video of no more than 5 minutes covering the project motivation, the dataset and audience, a demonstration of the final application, the principal design decisions, and the most important findings. The video should be accessible from the project page.

5.5 Poster

Prepare a poster for the Week 15 exhibition. The format may be creative, but the poster must clearly communicate the problem, data, visualization approach, key interactions, findings, and project access information. Include a QR code or short URL linking to the published project page or application.

5.6 Project page

The project page should include:

- project title and team members;
- representative image or visual abstract;
- concise abstract or project summary;
- link to the published application;
- link to the final paper;
- embedded or linked teaser and presentation videos;
- data sources and acknowledgments;
- instructions for running the software;
- links to the repository and any optional supporting material.

5.7 Repository, README, and code

The final repository must contain:

- all original source code required to build and run the visualization;
- data or documented data-acquisition instructions, subject to licensing and privacy restrictions;
- preprocessing scripts and reproducible instructions when applicable;
- the final paper and editable source;

- the poster and video links;
- a `README.md` with project overview, installation or execution instructions, deployment link, dependencies, data sources, and known limitations;
- a contribution statement describing how the work was divided among the team members;
- a brief reflection on the team's design and development process.

The teaching team should be able to follow the README and run the project without undocumented steps. Do not include credentials, private keys, or restricted data in the repository.

6 Evaluation Criteria

Work in Progress

Grades may be adjusted individually if repository history, contribution statements, peer evidence, or the ability to explain the work indicates substantially unequal participation.

7 General Quality and Submission Requirements

- All links must be tested before submission and remain accessible during evaluation.
- All visualizations must include appropriate titles, labels, legends, units, data sources, and explanations.
- Teams must acknowledge code, libraries, datasets, icons, images, and visual designs that influenced the work.
- The final application must not expose sensitive or personally identifiable information.
- The paper, poster, videos, page, and application must tell a consistent story and report the same claims.
- Every team member must be prepared to explain the data, code, visual encodings, interaction design, and findings.
- Late or incomplete submissions are subject to the course policies announced in the syllabus. A broken or inaccessible link is considered a missing component until corrected.

8 Social Activities Bonus

The following optional activity is intended to encourage social interaction beyond the technical work of the project.

Team members may earn bonus points if they participate in an **in-person social activity** with their team that is unrelated to coursework. Examples include an informal meal, a board or computer game, a puzzle or trivia activity, or a movie. To receive credit, submit evidence (until the Friday of week 15) that clearly shows the activity and the

participating team members. A selfie taken outside the classroom without a recognizable social activity does not qualify. Teams may join other teams for the activity.